

THREAT INTELLIGENCE ANALYST SIMULATION

Project — Tier-1 SOC Analyst Report
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Scenario

The SIEM fired an alert: workstation WS-FINANCE-04 made repeated outbound connections to 185.220.101.47 on port 443 at exactly 02:14 AM outside business hours. The session lasted 18 minutes and transferred 4.2 MB of data outbound. No user was logged in at the time.

Task 1: IOC Lookup & Classification

AlienVault OTX Lookup

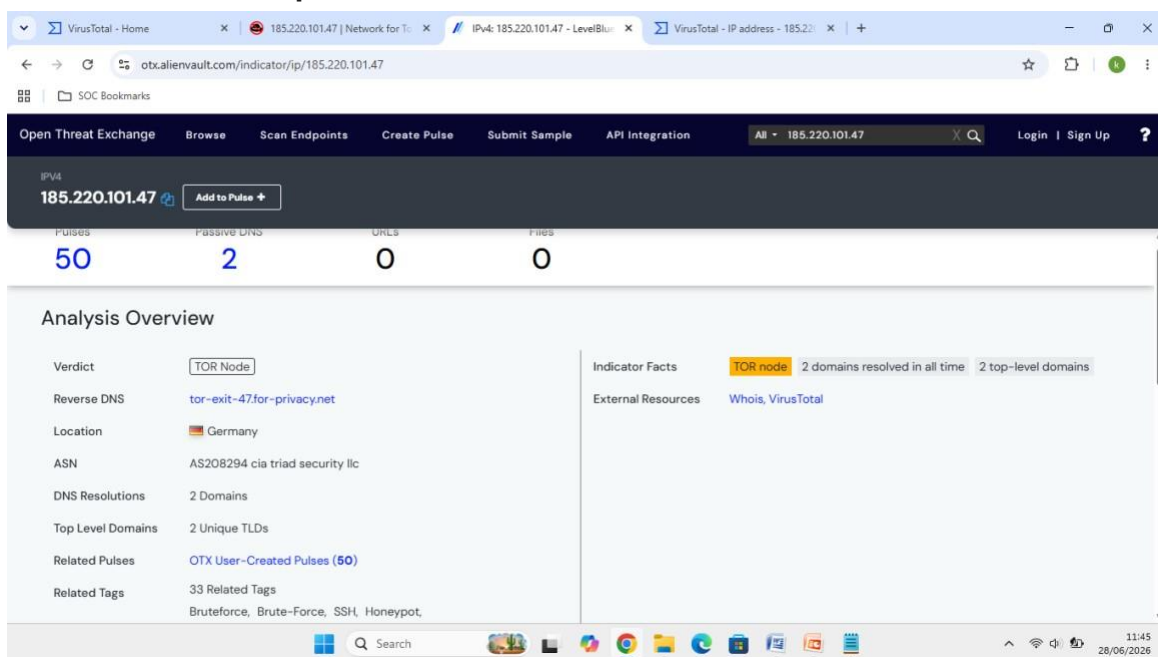


Figure 1: AlienVault OTX result for 185.220.101.47 — flagged as a Tor exit node, 50 related pulses.

AbuseIPDB Lookup

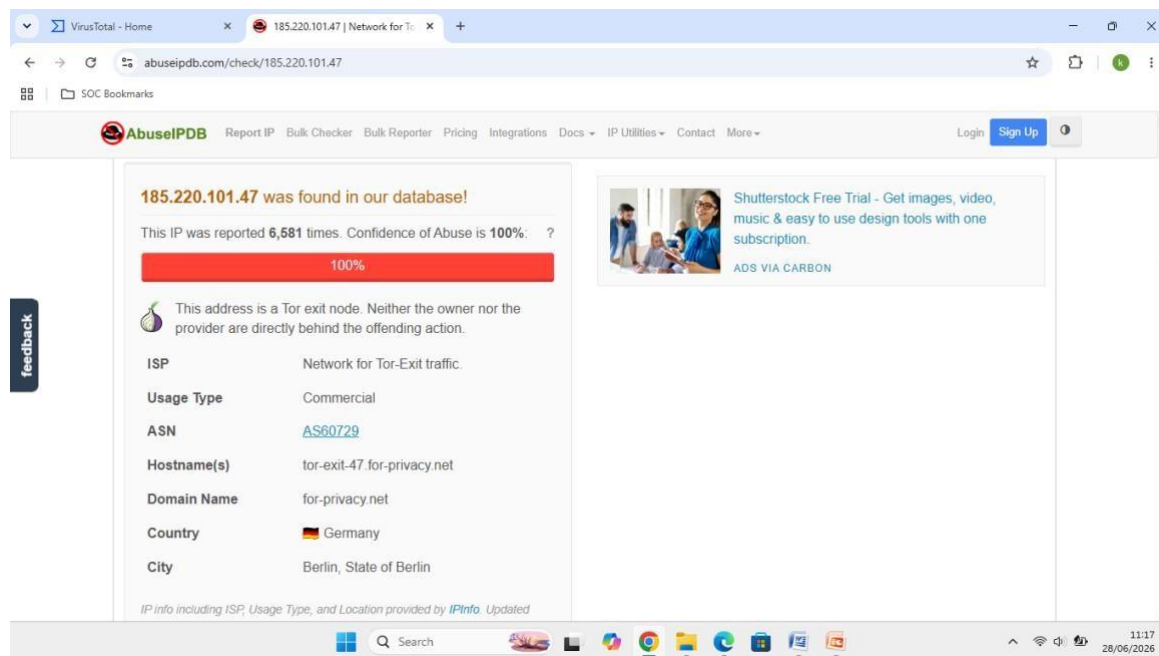


Figure 2: AbuseIPDB result for 185.220.101.47 — 100% abuse confidence, reported 6,581 times.

Findings

- AbuseIPDB: 185.220.101.47 has a 100% confidence of abuse score and has been reported 6,581 times. It is identified as a Tor exit node. ISP is listed as "Network for TorExit traffic," ASN AS60729, hostname tor exit 47.for-privacy.net, located in Berlin, Germany.
- AlienVault OTX: Verdict is TOR Node, reverse DNS tor exit 47.for-privacy.net, location Germany, ASN AS208294 (CIA Triad Security LLC), with 2 DNS resolutions across 2 top-level domains and 50 related pulses.
- VirusTotal: the IP has also been flagged as malicious, associated with malware and phishing activity.
- IOC type: this is a Network IOC. It is a malicious IP address, not tied to a file hash or file path (which would make it Host-Based), and it was not pulled from an email header (which would make it Email-Based).

Analysis

Both platforms independently confirm that 185.220.101.47 is a Tor exit node with a strong reputation for abuse 100% confidence on AbuseIPDB with 6,581 reports, and a TOR Node verdict on OTX backed by 50 related pulses. Tor exit nodes are commonly used to mask the true destination of traffic, so the fact that WS-FINANCE-04 reached out to this address outside business hours, with no user logged in, and pushed 4.2 MB outbound, strongly suggests this was C2 or data exfiltration traffic rather than legitimate user activity.

MITRE ATT&CK Mapping

Tactic	Technique ID	Technique Name	Justification
Initial Access	T1078	Valid Accounts	Adversaries may have used a compromised or stolen credential to access WS-FINANCE-04, which would explain a connection occurring with no user logged in.
Exfiltration	T1041	Exfiltration Over C2 Channel	The 4.2 MB outbound transfer over port 443 to a Tor exit node is consistent with data being exfiltrated through an existing command and control channel.
Exfiltration	T1048	Exfiltration Over Alternative Protocol	Routing the transfer through Tor (rather than a normal business application) suggests the use of an alternative protocol/channel chosen specifically to avoid standard monitoring.
Exfiltration	T1029	Scheduled Transfer	The connection occurred at a fixed, unusual time (02:14 AM) with no user present, suggesting the transfer was triggered on a schedule rather than by manual user action.
Execution	T1053	Scheduled Task/Job	The connection happened at a fixed time (02:14 AM) with no user logged in, suggesting a scheduled task may have triggered it automatically, letting the activity run unattended outside normal business hours.

Intel Type & Pyramid of Pain Analysis

This IOC sits at the Technical intelligence level. It is a single raw data point, an IP address that can be dropped straight into a block list or SIEM rule without further interpretation, which is the hallmark of technical intel rather than strategic, tactical, or operational intelligence.

On the Pyramid of Pain, an IP address sits at the Easy level, near the bottom of the pyramid. It is one of the cheapest indicators for an attacker to change.

Blocking IP is only a minor inconvenience to the attacker. Once it is blocked, the attacker can simply change to another IP which the firewall will not catch until that new IP is identified and blocked too.

A higher level IOC that would be more impactful to block is the attacker's Tactics, Techniques, and Procedures (TTPs), found at the top of the pyramid. Blocking based on TTPs forces the attacker to change their actual methodology, which is far more costly and time-consuming for them than switching IP addresses.

SIEM Correlation Rule & TI Feed Integration

Pseudo-code Correlation Rule

```
IF (destination_ip IN threat_intel_feed.tor_exit_nodes      OR destination_ip IN
threat_intel_feed.malicious_ips)      AND (connection_time BETWEEN 22:00 AND 06:00)
AND (logged_in_user == NULL)      AND (bytes_outbound > 1MB) THEN
trigger_alert(severity = HIGH, category = "Possible C2 / Exfiltration")
auto_action = "Isolate host, block destination IP"
```

TI Feed Integration Description

Integrate Virus Total directly into the SIEM as a live threat intelligence source. Indicators IPs, domains, and hashes are pulled on a scheduled basis and matched automatically against outbound connection logs. Any match against a known Tor exit node, C2 IP, or flagged malicious indicator generates an enriched alert in real time, instead of relying on an analyst to manually look up every suspicious IP after the fact.

Incident Summary Report

Executive Summary

The SIEM fired an alert after WS-FINANCE-04 made repeated outbound connections to 185.220.101.47 on port 443 at 02:14 AM outside business hours. The session lasted 18 minutes, transferred 4.2 MB of data outbound, and occurred with no user logged in to the workstation, a strong indicator that the connection was not legitimate user activity. Lookups against AbuseIPDB and AlienVault OTX confirmed the IP as a known Tor exit node with a 100% abuse confidence score and 6,581 abuse reports. Given the off hours timing, the absence of a logged-in user, and the volume of data moved, this activity is consistent with command-and-control communication or data exfiltration. I recommend the IP be blocked immediately, the host isolated for forensic review, and our detection logic updated to flag this pattern automatically.

IOC Findings

- IP Address: 185.220.101.47
- Abuse Score: 100% confidence of abuse on AbuseIPDB, with 6,581 historical abuse reports
- Threat Category: Confirmed Tor exit node (hostname tor-exit-47.for-privacy.net), also flagged by VirusTotal for association with malware and phishing activity
- Associated Campaigns: Linked to 50 community-reported pulses on AlienVault OTX; no single named campaign has been confirmed at this time

ATT&CK Mapping

Technique	Name
T1078	Valid Accounts
T1041	Exfiltration Over C2 Channel
T1048	Exfiltration Over Alternative Protocol
T1029	Scheduled Transfer
T1053	Schedule Task/Job

Containment Actions

- Block 185.220.101.47 at the perimeter firewall and add it to the threat intel block list immediately.
- Isolate WS-FINANCE-04 from the network and preserve memory/disk for forensic review to confirm or rule out compromise.

Long-Term Improvement

Integrate AlienVault OTX and AbuseIPDB directly into the SIEM as live threat intelligence feeds, with a correlation rule that auto-flags any outbound connection to a known Tor exit node or malicious IP that occurs outside business hours and with no logged-in user present. This closes the gap between an IOC being publicly known and it being detected on our network, instead of relying on an analyst to manually look up every suspicious IP after the fact. It would also let us catch similar activity on other finance-team workstations, rather than reacting one alert at a time.